Spectral–Spatial and Cascaded Multilayer Random Forests for Tree Species Classification in Airborne Hyperspectral Images

Fei Tong and Yun Zhang, *Senior Member, IEEE*

***Abstract*— The rapid development of remote sensing sensors has made it possible to collect airborne hyperspectral data with high spectral and spatial resolution. Such data can provide valuable information to identify tree species in the forest. However, it is a challenge to efficiently utilize the abundant spectral information and complex spatial information within the data. In this article, a spectral–spatial and cascaded multilayer random forests (SSCMRF) method is proposed to classify tree species in the high-spatial-resolution hyperspectral image. The SSCMRF adopts two classification stages to fully exploit the spatial information within shape-adaptive superpixels and shape-fixed patches. Two different kinds of spatial information are integrated by concatenating the output of the superpixel-based classification and the spectral features as the input of the patch-based classification. To demonstrate the superiority of the proposed SSCMRF, experiments are conducted with an airborne hyperspectral dataset of a forest area with a spatial resolution of 1 m. Training with 2.5% randomly selected ground-truth samples, the proposed SSCMRF achieves a classification accuracy of 97.50% within 6 min. In addition, the experiment results demonstrate that the proposed SSCMRF outperforms some state-of-the-art spectral–spatial classification models in terms of quantitative metrics and visual quality on the classification map.**

***Index Terms*— Airborne hyperspectral image, random forest (RF), spectral–spatial information, tree species classification.**

I. Introduction

SPATIAL explicit tree species composition information provides valuable information for a wide variety of applications including resource inventories, biodiversity assessment and monitoring, hazard and stress management, monitoring of invasive species, wildlife habitat mapping, and the overarching aim of a sustainable forest management [1]. However, manpower field survey for obtaining tree species information is inefficient, labor-intensive, and cost-effective [2]. Therefore, automatically extracting tree species information from remote sensing data is in practical demand. Hyperspectral data, which consists of hundreds of spectral bands, can provide abundant spectral information to discriminate different tree species in the forest. In addition to spectral information, spatial information contained in the hyperspectral data is also important. With higher spatial resolution, tree species having similar spectral characteristics can be distinguished according to spatial information. In recent years, the advancement of remote sensing sensors has made the acquisition of high-spatial-resolution hyperspectral data possible, especially for airborne data [3]. However, efficiently utilizing the abundant spectral and spatial information in the data for tree species classification is still a challenging task.

As different tree species have different spectral characteristics, some previous studies focused on the spectral information contained in the hyperspectral data [4]–[7]. Although hyperspectral data can provide abundant spectral information to discriminate tree species, the high-dimensional hyperspectral data also cause problems of increased data volume, a high correlation between bands, and reduced accuracy and efficiency of classification [2]. Feature selection and feature extraction methods have been proposed to reduce the dimension of the hyperspectral data and to extract the most discriminating bands or features [1], while it is still difficult to find a method that can be applied to all types of hyperspectral data. The support vector machine (SVM) [4] and random forest (RF) [8] are the two popular classifiers that have been used for classifying tree species using hyperspectral data [9]–[11]. However, the suboptimal choice of bands or features used in the classification may negatively affect the classification performance, and a limited number of training samples may not be enough for the training of SVM and RF [10]. Moreover, pixelwise classification methods that only use spectral information will waste the spatial information contained in the image and generate noisy classification maps, especially when the spatial resolution of the data is also high.

To fully exploit useful information from hyperspectral data, various spectral–spatial classification methods have been proposed in recent years. Generally, spatial information is exploited by assuming that the pixels within a local region usually represent the same material and have similar spectral characteristics [12]. One group of spectral–spatial methods focused on optimizing the pixelwise classification using the spatial information within local regions. In [13], pixelwise SVM classification result was optimized by conducting majority voting within segmented regions. Then, in [14], edge-preserving filters were applied to optimize the pixelwise

Manuscript received November 19, 2021; revised March 20, 2022; accepted May 18, 2022. Date of publication May 25, 2022; date of current version June 9, 2022. This work was supported in part by the Canadian Space Agency (CSA) under Grant 19FANBA36 and in part by the Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Program under Grant RGPIN-2021-03164. *(Corresponding author: Yun Zhang.)*

The authors are with the Department of Geodesy and Geomatics Engineering, University of New Brunswick, Fredericton, NB E3B 5A3, Canada (e-mail: ftong@unb.ca; yunzhang@unb.ca).

Digital Object Identifier 10.1109/TGRS.2022.3177935

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SVM classification result. Another group of spectral–spatial methods focused on conducting the local-region-based classification. In [15], spectral information and spatial information were integrated by measuring the similarity between image patches. In [16], spatial information within image patches was exploited by the joint sparse representation method. Then, in [17], shape-fixed image patches were replaced by oversegmented superpixels to exploit more accurate spatial information. In [18], local regions generated from the union of patches and corresponding superpixels were proposed to better exploit the spatial information. In [19], the SuperPCA was proposed to exploit spatial information by implementing the principal component analysis (PCA) [20] on local superpixels rather than on the entire image. The spatial information was integrated into the extracted PCA features to get more accurate classification results. Although a lot of spectral–spatial classification methods based on traditional machine learning methods have been proposed, most of them were validated on public datasets with low spatial resolution and small data volume. When dealing with high-spatial-resolution airborne hyperspectral data collected from the forestry area with strong spectral variability, obtaining accurate classification results is difficult.

As convolutional neural networks (CNNs) have got encouraging achievements in the field of computer vision [21], [22], some CNN-based methods were also proposed to exploit spectral–spatial information in hyperspectral images for classification in recent years. 2-D-CNN has been applied widely because a lot of computer vision tasks are focusing on processing natural images with RGB bands. For the task of hyperspectral image classification, although spectral information and spatial information can be integrated into 2-D convolution kernels, hundreds of channels in the hyperspectral image will lead to the overfitting problem, and the computation cost will be significantly increased because a large number of convolution kernels are required. To efficiently apply 2-D-CNN to hyperspectral classification, Yue *et al.* [23] applied PCA to project hundreds of spectral channels into a three-channel image for convolution. However, specific properties of the hyperspectral image might be wasted when using unsupervised and uncontrolled dimension reduction methods [24]. Then, in [25], 3-D-CNN was proposed to extract deep features from hyperspectral images. As structure features in the 3-D hyperspectral image could be fully utilized by 3-D kernels, the 3-D-CNN was able to efficiently exploit spectral–spatial information. In [26] and [27], the structures of 3-D-CNN were modified to optimize hyperspectral image classification. As for the tree species classification task, Zhang *et al.* [2] proposed a 3-D-CNN structure to fit for the high-spatial-resolution airborne hyperspectral data and achieved 95.74% classification accuracy. However, the training of a large number of parameters and the tuning of hyperparameters are still time-consuming. Although the 3-D–1-D CNN model that could shorten the training time of the 3-D-CNN model by 60% was also proposed, the classification accuracy decreased from 95.74% to 93.14%.

Beyond CNN-based deep learning methods, another kind of deep structure called deep forest was proposed in [28]. The deep forest is constructed by multiple levels of traditional machine learning classifiers. The classification results of the preceding level will concatenate original features as the input of the current level. The deep forest is able to generate comparable results as CNN-based methods and requires fewer hyperparameters. Inspired by deep forest, Cao *et al.* [29] proposed the densely connected deep RF (DCDRF) to exploit spectral–spatial information within image patches. In each layer of the DCDRF, each pixel in the patch will be classified by an RF, and the classification results of all pixels will be averaged as the output of this layer. However, spatial information in the image patch is not always correct. When the patch center is close to the edge of ground objects, pixels within the patch belong to different ground classes. Therefore, averaging classification results of all pixels in the patch may bring negative effects to classification performance.

According to the analysis of the existing literature, we propose a spectral–spatial and cascaded multilayer RFs' model called SSCMRF for tree species classification tasks using high-spatial-resolution airborne hyperspectral images. The proposed SSCMRF can efficiently make use of the spatial information within the data by integrating information from two kinds of spatial regions. Specifically, the contributions of this article are listed as follows: 1) by performing the classification based on multiscale superpixels using multiple RFs, the spatial information within different scales of superpixels is efficiently extracted; 2) by performing the classification based on shape-fixed patches using multilayer densely connected RFs, the spatial information within patches is fully utilized to optimize the classification result; 3) by concatenating the information extracted from multiscale superpixels and the original spectral features as the input for patch-based classification, the merits of both superpixel and patch are integrated to optimize the classification result; 4) by adopting a parallel implementation for the training of multiple RFs in each layer of the SSCMRF, the efficiency of the classification is obviously improved; and 5) the experiment results on a high-spatial-resolution hyperspectral airborne dataset demonstrate that the proposed methods outperform state-of-the-art spectral–spatial classification models proposed in recent years.

The rest of this article is organized as follows. In Section II, the details of the proposed method are presented. The experimental results are provided in Section III, and the discussions are provided in Section IV. In Section V, the conclusion of our work is presented.

II. Spectral–Spatial Classification Based on Cascaded Multilayer Random Forests

Spectral–spatial information in most of the published research is exploited with local regions. The shape-fixed patch and shape-adaptive superpixel are two kinds of local regions widely used in spectral–spatial classification models. To make full use of the spatial information contained in the high-spatial-resolution hyperspectral data, both the classification based on superpixels and the classification based on patches are adopted in the proposed SSCMRF model. The flowchart of the proposed SSCMRF model is shown in Fig. 1. The SSCMRF

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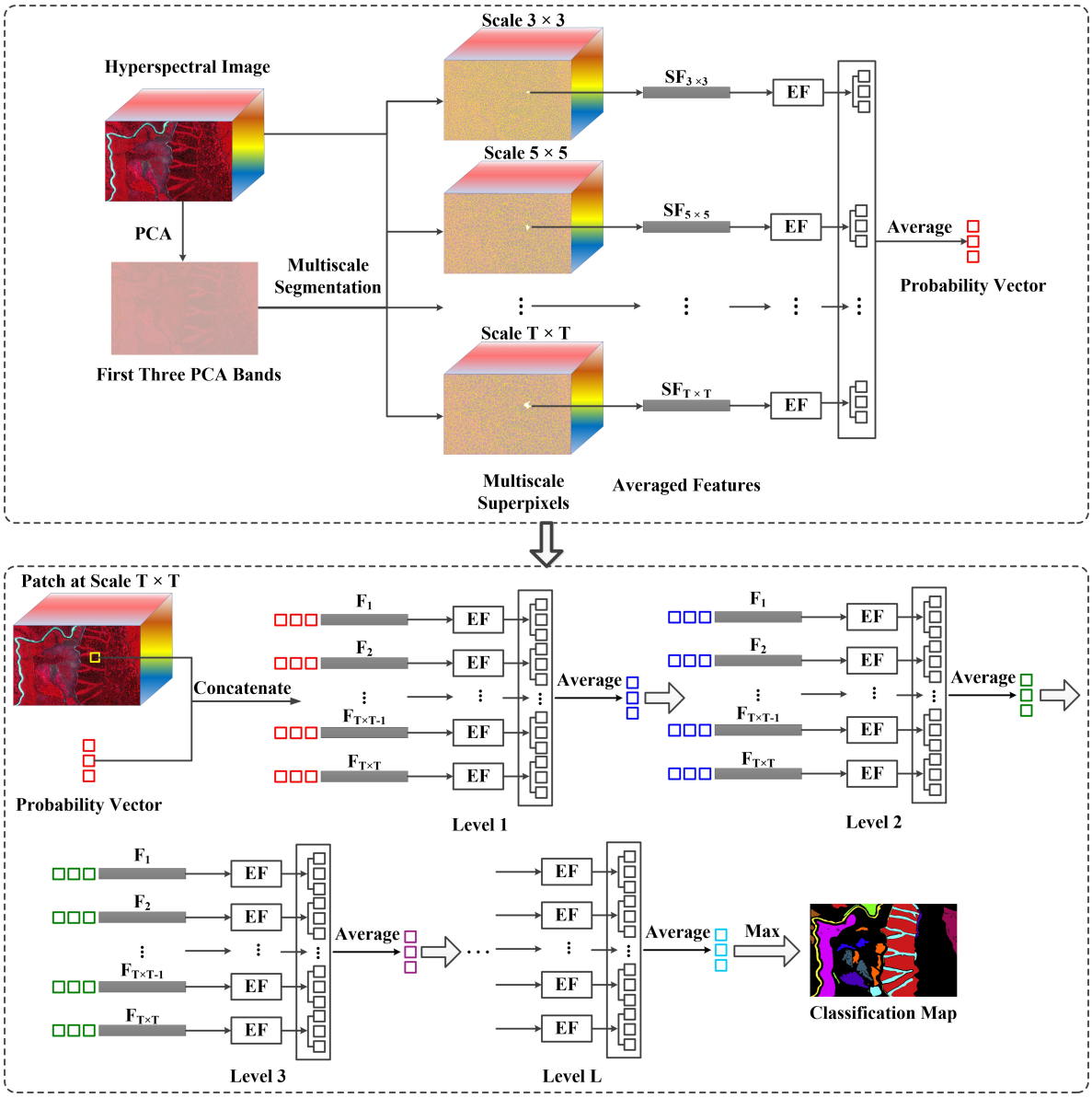


Fig. 1. Flowchart of the SSCMRF classification model.

model consists of two classification stages: 1) classification based on multiscale superpixels and 2) classification based on patches. The details of the two classification processes will be described in the following.

A. Classification Based on Multiscale Superpixels

As superpixels are shape-adaptive, and most of the boundaries of superpixels are adherent to ground objects' edges, the spatial information contained in superpixels is valuable for spectral–spatial classification. However, it is difficult to discriminate all ground objects using a single scale of superpixels because different ground objects have different spatial scales. Using a small spatial scale, superpixels tend to include pixels from a single kind of ground truth, while the small coverage of superpixels limits the exploitation of spatial information. Using a large spatial scale, spatial information is effectively exploited, while superpixels tend to include noise pixels, which negatively affects the classification performance. Therefore, in the first stage of the proposed model, spatial information is extracted from the classification based on multiscale superpixels. To generate multiscale superpixels, the entropy rate segmentation (ERS) [30] algorithm is adopted because of its high efficiency and accuracy [31]. To reduce the noise information contained in abundant spectral bands, the PCA [20] is applied to extract the first three components that contain most of the information. Then, ERS is executed on the first three PCA components using different spatial scales $(3 \times 3, 5 \times 5, \ldots, T \times T)$. As the influence of the number of PCA components on the final classification result is slight, users can determine the number of PCA components according to their preference. The largest spatial scale for superpixels is

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$T \times T$, which requires to be assigned by users. Assuming that the size of the hyperspectral image is $W \times H$, the number of superpixels $\mathrm{Num}_{t\times t}$ for the scale $t \times t$ is

$$\mathrm{Num}_{t\times t} = \frac{W \times H}{t \times t}. \tag{1}$$

After multiscale segmentation, $(T-1)/2$ scales of superpixels are generated. For arbitrary pixel P_i in the image, multiscale superpixels that include P_i are $S^i_{3\times3}, S^i_{5\times5}, \ldots, S^i_{T\times T}$. The features of multiscale superpixels are $SF^i_{3\times3}$, $SF^i_{5\times5}, \ldots, SF^i_{T\times T}$. For arbitrary scale $t \times t$, the superpixel features $SF^i_{t\times t}$ are the average spectral features of all pixels within the superpixel $S^i_{t\times t}$. Multiple extremely RFs (EFs) [32] are adopted to classify different scales of superpixels. In the training process, the class probability vector generated by each EF is produced by fivefold cross-validation to avoid overfitting. The outputs of EFs are class probability vectors $V^i_{3\times3}, V^i_{5\times5}, \ldots, V^i_{T\times T}$. Finally, to integrate information from different spatial scales, class probability vectors generated from different scales of superpixels are averaged to be one probability vector V_i

$$V_i = \frac{V^i_{3\times3} + V^i_{5\times5} + \cdots + V^i_{T\times T}}{(T-1)/2}. \tag{2}$$

So far, for arbitrary pixel P_i, one class probability vector V_i is extracted from the classification based on multiscale superpixels.

B. Classification Based on Patches

In the second stage of classification, spatial information within patches is exploited by a densely connected multilayer RFs' structure [29]. The basic classifier used in [29] is the RF classifier, while the proposed SSCMRF chooses to use the EF classifier because the EF classifier can efficiently produce more robust classification results [33]. For arbitrary pixel P_i in the image, assuming that PT_i represents the patch centered by P_i, in each layer, all pixels within PT_i will be classified by multiple EFs, and all classification results will be averaged to be the output of P_i. The average operation can make use of the spatial information within patches under the assumption that pixels belonging to local patches tend to have similar spectral characteristics. However, when P_i is close to the boundaries of ground objects, PT_i tends to include pixels from different ground-truth classes. Thus, the spectral characteristics of some noise pixels in PT_i are different from that of P_i. If the classification is based on original spectral features, the classification results of these noise pixels tend to be different from that of P_i. Therefore, the output of P_i may be misled by the average operation. To solve this problem, we propose to concatenate normalized values of hundreds of spectral bands with the class probability vector V_i obtained from the first stage of classification. The input features for each pixel in PT_i are the concatenation of its spectral features and V_i. In this way, the class probability vector for P_i extracted from multiscale superpixels can affect the classification results of all pixels in the PT_i. Even for noise pixels having different spectral characteristics from P_i, the classification results can be similar to that of P_i. In this way, the shortage of utilizing spatial information in the patch is able to be made up for by the spatial information extracted from multiscale superpixels.

The size of the patch is $T \times T$, which is as same as the largest scale of superpixels. Each patch will contain T^2 pixels. Assuming that there are m training samples, expanded by local patches, $m \times T^2$ training pixels will be fed into the densely connected multilayer RFs' model. In level 1, concatenated features of $m \times T^2$ training pixels are fed into T^2 EFs, respectively, for training. Then, for each training pixel, a classification probability vector will be produced by fivefold cross-validation to reduce the risk of overfitting. Finally, class probability vectors produced by T^2 EFs are averaged to be the output of level 1. By averaging the classification results of all pixels in the patch, the model becomes more robust. In subsequent levels, the input is the concatenation of the preceding level's output and original normalized spectral features, and the training process is as same as that in level 1. The cascaded multilayer training process will stop until the number of cascade levels reaches the preassigned maximum levels L. In the testing process, the output of the last level L is the class probability vector V^L_i. For arbitrary pixel P_i, the final label is the label of the class with the highest probability in V^L_i.

C. Parallel Optimization

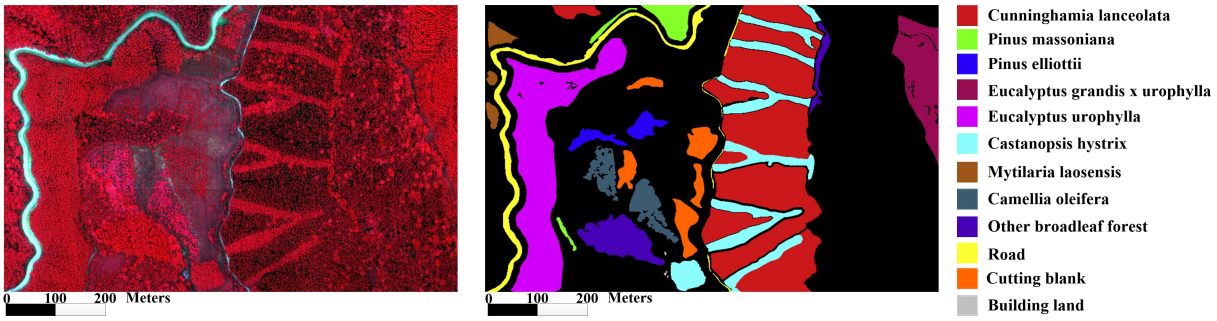
The spatial information in patches and multiscale superpixels is well exploited to discriminate tree species in the proposed SSCMRF. However, when the spatial scale $T \times T$ is large, the number of pixels in the patch is notable. A lot of EFs will be used in each layer of the patch-based classification. In the previous research [29], no optimization was applied to speed up the classification process. When the spatial scale $T \times T$ becomes large, the time cost increases dramatically. Therefore, in the proposed model, a parallel scheme is adopted to speed up the classification process. The classifications of multiple EFs in the same layer are executed parallel using the joblib library [34] in Python. In this way, the computing resource of the computer is fully utilized. The reduction of the time cost brought by the parallel optimization is affected by the number of cores in the computer's CPU.

III. Experiments

A. Dataset

To validate the performance of the proposed SSCMRF, the airborne hyperspectral dataset used in [2] was adopted in experiments. The hyperspectral data were captured on January 13 and January 30, 2018, from a research area located at the Jiepai branch of the Gaofeng State Owned Forest Farm in Guangxi province, South China (22°12′36″–22°2′24″N, 107°19′40″–109°37′3″E). The AISA Eagle II diffraction grating push-broom hyperspectral imager carried by the airborne LiCHy (light detection and ranging (LiDAR), charge-coupled device (CCD), and Hyperspectral) system of the Chinese Academy of Forestry Sciences was used for collecting the airborne hyperspectral data. The detailed parameter indicators of the hyperspectral dataset sensor can be found in [2]. The hyperspectral data have 125 spectral bands ranging from 400 to 1000 nm with a spectral resolution of 3.3 nm. The

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(a) (b)

Fig. 2. (a) False color image of the hyperspectral data. (b) Ground-truth classification map.

TABLE I
NUMBER OF GROUND-TRUTH PIXELS FOR ALL CATEGORIES

Class	Training	Validation	Testing
Cunninghamia lanceolata	2019	2019	76705
Pinus massoniana	134	134	5072
Pinus elliottii	104	104	3967
Eucalyptus grandis x urophylla	373	373	14185
Eucalyptus urophylla	1007	1007	38261
Castanopsis hystrix	637	637	24187
Mytilaria laosensis	75	75	2850
Camellia oleifera	208	208	7908
Other broadleaf forest	269	269	10203
Road	262	262	9948
Cutting blank	254	254	9662
Building land	5	5	58
Total	5347	5347	203006

spatial resolution of the data is 1 m, and the size of the data is 906×572 pixels. A total of 213 700 samples of nine forest vegetation categories and three nonforest vegetation categories are labeled in this dataset. The false color image of the hyperspectral data [red: infrared (837.19 nm); green: red (687.73 nm); and blue: green (541.40 nm)] and the ground-truth classification map are shown in Fig. 2. In the classification, 2.5% of ground truth was randomly selected for training. As fivefold cross-validation was adopted in the training process of the proposed model, the minimum number of training samples for each class was set as 5. The size of the validation set was as same as that of the training set. The rest 95% of the ground truth was the testing set. The numbers of ground-truth pixels for all categories are tabulated in Table I.

B. Classification Performance

To validate the performance of the proposed SSCMRF model, experiments were conducted on the selected dataset using different classification methods for comparison. Classification accuracy of each class, overall accuracy (OA), average accuracy (AA), and kappa coefficient were adopted to evaluate the classification accuracy. All classification results were averaged by ten runs with different random training samples to make the comparison more reliable. For comparison, the extended attribute profiles (EAPs) [35] and invariant attribute profiles (IAPs) [36] that focus on exploiting spatial information were adopted to extract features from hyperspectral data, and then, extracted features were classified using the SVM classifier. EAPs were extracted from the first four PCA components, and other parameters used to extract EAPs were as same as that used in [35]. Validated by the validation set, the parameters used to extract IAPs were set as $n_s = 3$, $r = [2, 4, 6]$, $m = [0, 1, 2, 3]$, $d = 50$, and $n_g = 5$. The optimal parameters of the SVM were set by a grid search to achieve the best accuracy on the validation set. Two superpixel-based classification models were implemented for comparison. The first one is the SuperPCA, and the second one is the multiscale extension of SuperPCA, which is called MSuperPCA [19]. Validated by the validation set, the number of superpixels used in SuperPCA was set as 100. For MSuperPCA, the fundamental superpixel number was set as 100, and the scale parameter C was set as 4. The number of PCA features extracted in both SuperPCA and MSuperPCA was set as 30. The optimal parameters of the SVM used in SuperPCA and MSuperPCA were set by a grid search to achieve the best accuracy on the validation set. The 3-D-CNN proposed in [2] was also executed for comparison, and the hyperparameters used in 3-D-CNN were as same as that used in [2]. The light version 3-D–1-D CNN was not considered because its accuracy was expected to be lower than that of 3-D-CNN. Except for 3-D-CNN, the DCDRF that uses multilayer densely connected RFs [29] was implemented for comparison. The tree number in each forest of the DCDRF and the proposed SSCMRF was set as 5 because it has been proven in [29] that five trees are enough to get satisfactory results. The spatial scale $T \times T$ used in DCDRF and the proposed SSCMRF was 13×13. The number of cascaded layers for DCDRF was set as 6, while the number of cascaded layers L used in the proposed SSCMRF was 3. The classification results are tabulated in Table II. The classification maps generated from one of ten runs are shown in Fig. 3.

From the classification result, it can be found that, although spatial information was utilized to extract EAPs, the classification result still had a lot of misclassifications. The classification based on IAPs performed better than that based on EAPs because features in IAPs were extracted in both spatial and frequency domains. Compared with SuperPCA, as MSuperPCA utilized spatial information in multiscale superpixels, the classification accuracy of MSuperPCA was higher than that of SuperPCA. However, the classification accuracy was still far

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TABLE II
CLASSIFICATION METRICS (AVERAGED ON TEN RUNS) OF ALL METHODS

Class	EAPs	IAPs	SuperPCA	MSuperPCA	3D-CNN	DCDRF	SSCMRF
Cunninghamia lanceolata	94.81	95.35	93.25	**97.25**	95.53	94.92	96.66
Pinus massoniana	69.69	81.80	90.66	91.35	94.84	83.02	**98.72**
Pinus elliottii	41.20	82.28	92.07	92.99	91.42	79.49	**99.60**
Eucalyptus grandis x urophylla	92.99	94.23	99.32	99.67	97.56	97.71	**99.94**
Eucalyptus urophylla	94.61	96.15	98.68	99.43	97.50	97.30	**99.61**
Castanopsis hystrix	78.62	77.58	45.52	47.32	88.92	85.34	**92.12**
Mytilaria laosensis	85.80	90.01	97.34	97.82	92.04	92.89	**98.73**
Camellia oleifera	67.44	93.72	87.39	93.55	96.40	96.81	**98.85**
Other broadleaf forest	70.66	81.51	90.40	88.69	93.85	96.40	**99.06**
Road	91.01	98.22	95.51	96.22	98.76	98.28	**98.96**
Cutting blank	75.29	97.82	97.81	98.55	98.13	99.70	**99.72**
Building land	45.52	74.31	74.66	77.76	69.28	50	**100**
OA(%)	87.51	92.13	88.93	91.09	95.35	94.31	**97.50**
AA(%)	75.64	88.58	88.55	90.05	92.85	89.32	**98.50**
Kappa×100	83.97	89.99	85.82	88.52	94.12	92.78	**96.84**

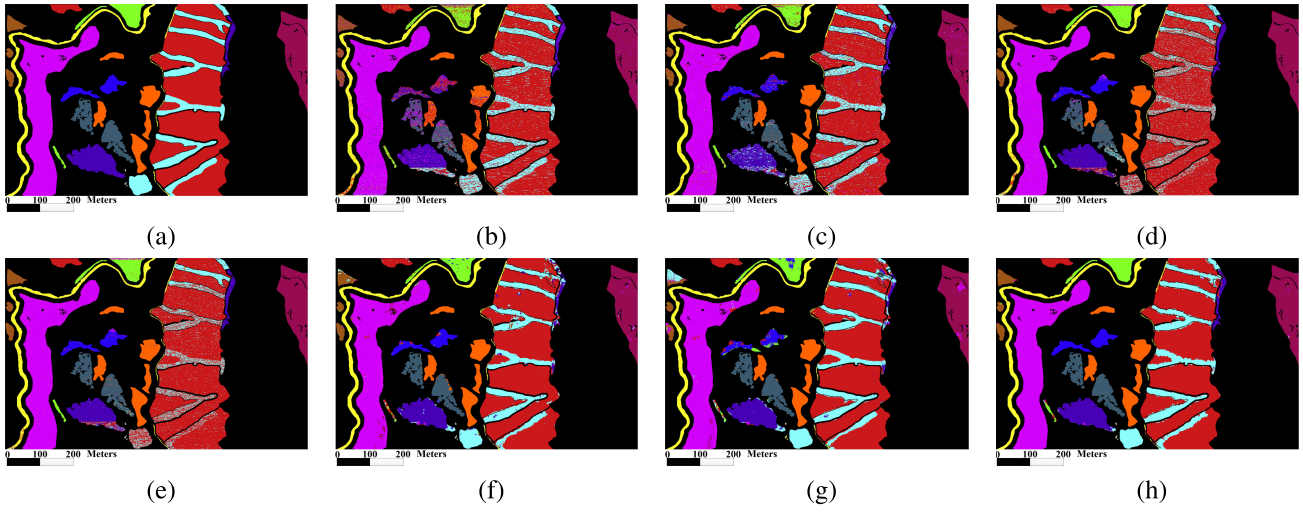


Fig. 3. Classification maps generated by different methods: (a) reference, (b) EAPs (OA = 87.81%), (c) IAPs (OA = 92.47%), (d) SuperPCA (OA = 89.33%), (e) MSuperPCA (OA = 91.22%), (f) 3-D-CNN (OA = 95.54%), (g) DCDRF (OA = 94.39%), and (h) SSCMRF (OA = 97.72%).

from satisfactory. Especially for the tree species Castanopsis hystrix, the accuracies of SuperPCA and MSuperPCA were lower than 50%. Besides, it can be found that, although both 3-D-CNN and DCDRF exploited spatial information within shape-fixed patches, the 3-D-CNN outperformed the DCDRF, especially for classes with a small number of training samples, such as Pinus massoniana and Pinus elliottii. The proposed SSCMRF generated the best classification result among all methods. Compared with the 3-D-CNN model, the increases in three evaluation metrics achieved by the proposed SSCMRF were 2.15% for OA, 5.65% for AA, and 2.72 for kappa × 100. In nine tree species classes, except for tree species Cunninghamia lanceolata, the accuracy of SSCMRF on the other eight tree species was the highest among all methods. Especially for tree species Pinus massoniana and Pinus elliottii, the SSCMRF improved the accuracy by 3.88% and 6.61% compared to the highest accuracy of other methods. For tree species Castanopsis hystrix, although the SSCMRF achieved the highest accuracy of 92.12% among all methods, 92.12% is relatively low compared to the accuracy of other tree species. The classification accuracy of the tree species Castanopsis hystrix was negatively affected by the tree species Cunninghamia lanceolata because the locations of these two tree species are very close in the image.

From the classification maps, it can be found that classification results of EAPs, IAPs, SuperPCA, and MSuperPCA had a lot of salt and pepper noises. In the classification map of the 3-D-CNN, most of the misclassifications were close to the edge of ground objects because patches close to the edge of ground objects contained a lot of noise pixels that would negatively affect the classification accuracy of 3-D-CNN. In addition, some obvious misclassifications were still found inside the ground objects. The classification map of the DCDRF shows a similar pattern like that of the 3-D-CNN because DCDRF also used patches to exploit spatial information. For both 3-D-CNN and DCDRF, a lot of misclassifications were found in similar positions, which means that patch-based classification models could not correctly classify these pixels. In the classification map of the proposed SSCMRF, although misclassifications still could be found near the edge of ground

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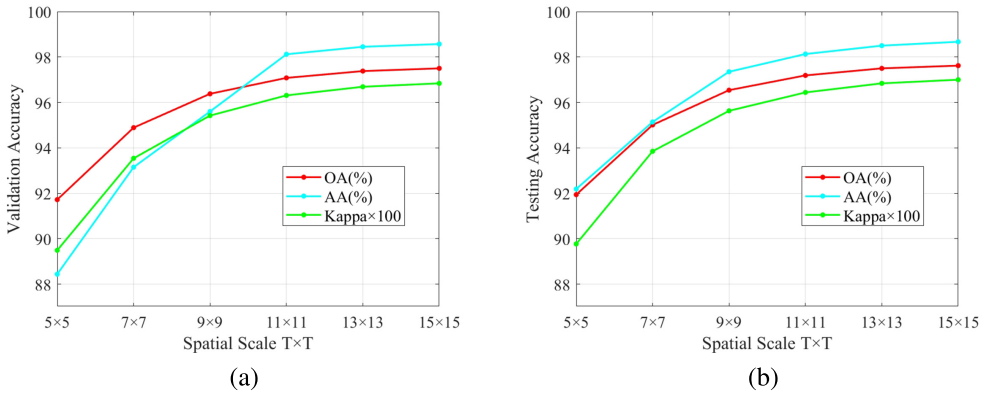


Fig. 4. Effect of spatial scale on classification metrics: (a) validation set and (b) testing set.

objects, the misclassifications were reduced compared to 3-D-CNN and DCDRF. In addition, misclassifications inside the ground objects were greatly reduced in the classification map of the SSCMRF.

IV. Discussion

To further verify the effectiveness of the proposed SSCMRF model, multiform qualitative and quantitative analyses are conducted in this section. The analyses cover the influence of two hyperparameters on the classification accuracy, impact of concatenation, small-sample learning ability, running efficiency, and the performance on the benchmark dataset. All experimental results shown in this section are the average of ten runs. The configuration of all methods is as same as that used in Section III unless the changes are mentioned in this section.

A. Effect of Spatial Scale on Classification Accuracy

Spatial scale is a very important hyperparameter for the spectral–spatial model because it determines the size of the spatial receptive field. In the proposed SSCMRF, the spatial scale $T \times T$ determines how many scales are applied to produce multiscale superpixels and how many EFs are used in each layer of the patch-based classification. The memory cost and efficiency of the proposed SSCMRF are also indirectly affected. To evaluate the impact of spatial scale on the performance of the proposed SSCMRF, classifications were conducted using different spatial scales (5×5, 7×7, 9×9, 11×11, 13×13, and 15×15). The classification results on the validation set and the testing set are shown in Fig. 4.

On both the validation set and the testing set, better classification accuracy was achieved by using a larger spatial scale because more spatial information was exploited. However, when the spatial scale is too large, the large superpixels and patches cannot provide useful information because more noises are also included. As shown in Fig. 4(a), when the spatial scale reached 13×13, increasing the spatial scale to 15×15 could only bring a slight improvement in accuracy. However, the number of EFs used in each layer of the patch-based classification increased from 169 to 225, and the input of the patch-based classification contained more training samples. The time cost and the memory cost of the proposed SSCMRF would increase obviously. Therefore, based on the results generated from the validation set, the optimal spatial scale was 13×13. The results on the testing set [see Fig. 4(b)] also showed the same pattern, which means that the hyperparameter $T \times T$ obtained from the validation set perfectly fit the testing set.

B. Effect of Cascade Level Number on Classification Accuracy

The number of cascade level L used in patch-based classification is another hyperparameter that affects the performance of the proposed SSCMRF method. Therefore, classifications using different numbers of cascade levels (ranging from 1 to 8) were conducted to show the impact of the hyperparameter L. The classification results on the validation set and the testing set are shown in Fig. 5.

For the validation set, all classification metrics were increasing obviously in the first few layers. Especially, the biggest performance increase occurred when the cascade level L increased to 2. When the cascade level L reached 3, increasing the number of cascade levels was not helpful for improving the classification performance, while the time cost would increase. Therefore, the optimal number of cascade levels was 3 for the validation set. For the testing set, increasing the number of cascade levels did not change the AA. In addition, when the cascade level reached 3, increasing the number of cascade levels will not improve the OA and kappa obviously. Therefore, for both the validation set and the testing set, the optimal number of cascade levels L is 3.

C. Impact of Concatenation

Concatenating superpixel-based classification results with the spectral features as the input of patch-based classification plays an important role in optimizing the classification model. To validate the impact of concatenation operation, the patch-based classification that only uses spectral features as input was implemented for comparison. The classification results are listed in Table III. The numbers of misclassified internal pixels and misclassified edge pixels were counted to show the effect of concatenation. In the ground-truth map, if all

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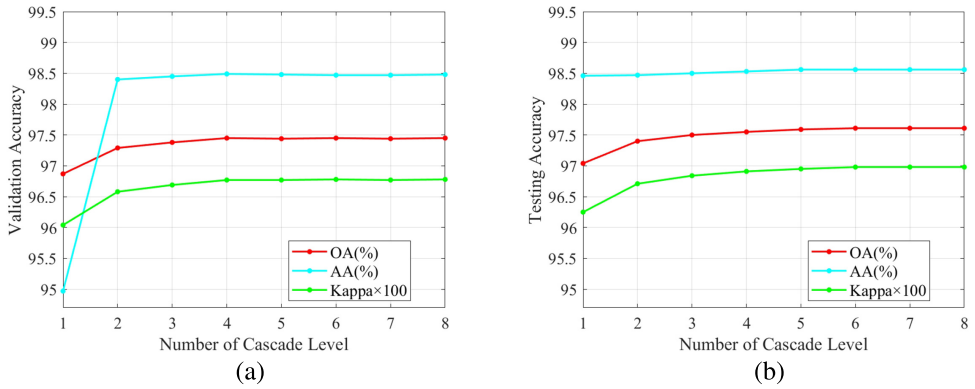


Fig. 5. Effect of cascade level number on classification metrics: (a) validation set and (b) testing set.

TABLE III
IMPACT OF CONCATENATION ON CLASSIFICATION PERFORMANCE

Metrics	Patch-based Classification	SSCMRF
Misclassified Internal Pixels	1765	89
Misclassified Edge Pixels	9782	4985
OA(%)	94.31	97.50
AA(%)	93.82	98.50
Kappa*100	92.78	96.84

pixels in the patch have the same class label, the patch center is the internal pixel. For each edge pixel, the patch centered by it contains pixels from multiple classes. Compared with internal pixels, edge pixels were more likely to be misclassified by both two methods. The proposed model outperformed the patch-based classification for all metrics, and the accuracy increase was more than 3%. After applying the concatenation operation, both the numbers of misclassified internal pixels and misclassified edge pixels were reduced. Most of the internal pixels misclassified in patch-based classification were correctly classified after concatenating superpixel-based classification results because the spatial information contained in multiscale superpixels was helpful for improving classification performance. The misclassified edge pixels are difficult to be corrected because noise pixels are included in the patch. However, after concatenation, about half of the edge pixels misclassified in patch-based classification were correctly classified. It proves that the concatenation operation is effective to remove the negative influence caused by noise pixels in the patch.

D. Small-Sample Learning Ability

Given enough amount of training samples, most of the existing methods are able to achieve satisfactory performance. However, it is not easy to get enough amount of training samples because obtaining labeled samples is labor-intensive and cost-effective in most practical applications. Therefore, the performance of the classification model using a small number of training samples is worth to be evaluated. Specifically, 1%, 0.5%, 0.2%, and 0.1% of the ground truth were randomly selected as training samples. The minimum number of training samples for each class was set as 5. The OA indices obtained by different methods are compared in Table IV.

As shown in Table IV, the proposed SSCMRF outperformed all of the other methods using different percentages of training samples. When only 1% of the ground truth were randomly selected as training samples, the classification accuracy of the proposed SSCMRF achieved 95.80%. When the percentage of training samples was 0.1%, the OA of the proposed SSCMRF was at least 5% higher than that of other methods. When the percentage of training samples was lower than 0.5%, the performance of the 3-D-CNN was the poorest because the CNN-based deep learning model required enough training samples to optimize the whole network. These results strongly prove the outstanding small-sample learning ability of the proposed SSCMRF.

E. Comparison of Running Efficiency

Efficiency is another important metric for evaluating the performance of the classification methods. In this section, the computation efficiency of all classification methods is investigated. All experiments were conducted on a desktop with Intel[1] Core[2] i7-2600 CPU @ 3.40 GHz and NVIDIA GeForce GTX 970 GPU. EAPs, IAPs, SuperPCA, and MSuperPCA were executed on MATLAB R2019b. 3-D-CNN, DCDRF, and the proposed SSCMRF were executed on Python 3.6.8. Both CPU and GPU were utilized by 3-D-CNN, while other methods only used CPU. To show the improvement brought by the parallel scheme proposed in this article, the time costs of DCDRF and SSCMRF without parallel optimization were also recorded. The time costs for all classification methods are tabulated in Table V.

Although SuperPCA, MSuperPCA, and the proposed SSCMRF cost extra time for segmentation, the time cost of segmentation was slight compared to the total time cost. The SuperPCA was the most efficient method, while its classification accuracy was not satisfactory. The time cost of the 3-D-CNN was the highest among all methods because the training process required too much time. Using the parallel optimization, the proposed SSCMRF was more than two times

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TABLE IV
OA (%) OBTAINED FROM DIFFERENT NUMBERS OF TRAINING SAMPLES

Percentage	EAPs	IAPs	SuperPCA	MSuperPCA	3D-CNN	DCDRF	SSCMRF
0.1%	68.88	76.68	76.84	79.31	64.95	82.14	**87.71**
0.2%	75.40	81.00	83.22	85.73	75.15	87.16	**90.53**
0.5%	81.19	86.64	86.60	88.63	88.61	90.87	**94.28**
1%	84.06	89.61	87.78	89.65	92.63	92.64	**95.80**

TABLE V
TIME COST (S) OF DIFFERENT METHODS

	EAPs	IAPs	SuperPCA	MSuperPCA	3D-CNN	DCDRF	Parallel DCDRF	SSCMRF	Parallel SSCMRF
Segmentation	-	-	6	47	-	-	-	37	37
Feature Extraction	1004	121	2	11	-	-	-	-	-
Training	29	11	12	103	2196	1133	366	227	105
Testing	91	46	45	412	85	579	403	320	168
Total	1124	178	65	573	2281	1712	769	584	310

TABLE VI
CLASSIFICATION METRICS (AVERAGED ON TEN RUNS) OF ALL METHODS ON THE UNIVERSITY OF PAVIA DATASET

Class	EAPs	IAPs	SuperPCA	MSuperPCA	3D-CNN	DCDRF	SSCMRF
Asphalt	**99.41**	98.20	92.47	97.99	91.89	97.02	98.44
Bare soil	**99.85**	99.65	99.21	99.75	93.18	99.24	99.63
Bitumen	97.32	85.81	97.90	97.90	81.01	96.04	**99.01**
Bricks	97.91	97.62	89.23	95.11	98.39	97.89	**99.21**
Gravel	99.61	99.75	96.13	97.45	**99.86**	99.47	99.49
Meadows	**99.55**	98.56	96.61	98.03	86.81	97.36	99.25
Metal sheets	99.41	94.51	92.84	95.07	84.19	99.02	**99.96**
Shadows	**98.70**	93.00	93.43	97.74	84.45	96.84	97.20
Trees	99.71	98.23	98.35	99.29	96.78	99.77	**99.87**
OA(%)	**99.36**	97.71	96.27	98.45	91.28	98.21	99.13
AA(%)	99.00	96.15	95.13	97.59	90.73	98.07	**99.12**
Kappa×100	**99.15**	96.97	95.05	97.95	88.32	97.63	98.84

faster than the DCDRF because EFs used in SSCMRF are more efficient than RFs used in DCDRF, and SSCMRF only requires three cascade levels in patch-based classification, while DCDRF requires six cascade levels. It needs to be noted that the total time cost of the proposed SSCMRF was shorter than 6 min, while the classification accuracy could achieve 97.50%. If the parallel optimization was not applied in DCDRF, the time costs for training and testing were 1133 and 579 s. Moreover, without the parallel optimization in the proposed SSCMRF, the time costs for training and testing were 227 and 320 s. It proves that the parallel scheme adopted by the proposed method is effective to improve the classification efficiency.

F. Performance on Benchmark Dataset

To further validate the performance of the proposed SSCMRF method, experiments were conducted on the University of Pavia dataset with a high spatial resolution of 1.3 m. It contains 103 spectral bands with the size of 610 × 340 pixels. The spectral coverage of the University of Pavia dataset is from 0.43 to 0.86 μm. There are 42 776 labeled samples and nine classes on the ground-truth map. In experiments, 2.5% of the labeled samples were randomly selected as training samples. The size of the validation set was as same as that of the training set. Parameters of all methods were tuned to get optimal accuracy on the validation set. n_g in IAPs was set as 30. The number of superpixels used in SuperPCA was 20. For MSuperPCA, the fundamental superpixel number was set as 20, and the scale parameter C was set as 6. The size of the neighborhood used in 3-D-CNN was set as 13 × 13. The classification results obtained by different methods are compared in Table VI. As can be observed in Table VI, except for 3-D-CNN that requires more training samples, other methods can get an accuracy higher than 95%. The classification based on EAPs got the highest accuracy 99.36%. Although the accuracy of the proposed SSCMRF is a little lower (0.23%) than that of the classification based on EAPs, the accuracy of the proposed SSCMRF is higher than 99%. Therefore, the proposed SSCMRF still shows a high ability to discriminate ground classes in the benchmark dataset.

V. CONCLUSION

In order to improve the tree species classification accuracy and efficiency, this article proposes a spectral–spatial model to fully utilize the available information in high-spatial-resolution

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hyperspectral data. Two classification stages are designed to exploit and integrate spatial information in patches and multiscale superpixels. In the patch-based classification stage, the proposed SSCMRF adopts a similar model structure as the DCDRF. However, the idea of combining two kinds of spatial information is really helpful to optimize classification performance. The proposed SSCMRF is not only able to achieve higher classification accuracy but also significantly accelerate the speed of convergence because it only requires three levels of classifications to achieve satisfactory results, while the DCDRF requires six levels of classifications. It also explains why the time cost of the proposed SSCMRF is obviously lower than that of the DCDRF.

Although the 3-D-CNN model can also achieve a high classification accuracy (95.35%), it is still difficult for normal users to achieve a satisfactory result because too many hyperparameters are required to be assigned by users. The time cost of the training process is expensive, and even the GPU is utilized for acceleration. Therefore, tuning hyperparameters is difficult and time-consuming. However, the proposed SSCMRF achieved the best classification performance (97.50%), and it only required two hyperparameters that need to be tuned by users. In addition, the training efficiency of the SSCMRF is high, which means that the process of searching for the optimal hyperparameters is also efficient. Therefore, from the perspective of practical application, the proposed SSCMRF has a high potential to be used as a basic classification tool by researchers who are interested in forest management applications.

ACKNOWLEDGMENT

The authors would like to thank Dr. Bin Zhang for providing the dataset and the source code of the 3-D-convolutional neural network (CNN) model on Github (https:// github.com/ Bin-Zh/ Three-dimensional-convolutional-neural-network-model-for-tree-species-classification-using-airborne-).

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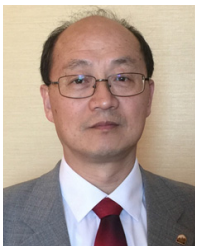
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Fei Tong received the B.S. and M.S. degrees in computer science and technology from the China University of Geosciences, Wuhan, China, in 2015 and 2018, respectively. He is currently pursuing the Ph.D. degree with the University of New Brunswick, Fredericton, NB, Canada.

His main research interests include remote sensing image processing and artificial intelligence.



Yun Zhang (Senior Member, IEEE) received the Ph.D. degree from the Free University of Berlin, Berlin, Germany, in 1997.

He was a Canada Research Chair in advanced geomatics image processing and a Visiting Professor with the Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, and a Research Scientist with the German Aerospace Center (DLR), Berlin. He is currently a Professor of remote sensing with the University of New Brunswick, Fredericton, NB, Canada. He has authored or coauthored over 260 research papers. The research outcomes from his laboratory have resulted in 15 patents, three new patent applications, and more than ten technology licenses. Users of the technologies include NASA, Google, the Natural Resources of Canada, DigitalGlobe, and many others.

Dr. Zhang is also a fellow of the Canadian Academy of Engineering (FCAE). He received many prestigious national and international research awards.

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